Day 2 : Power Supply Systems Theory

Task 1 : List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

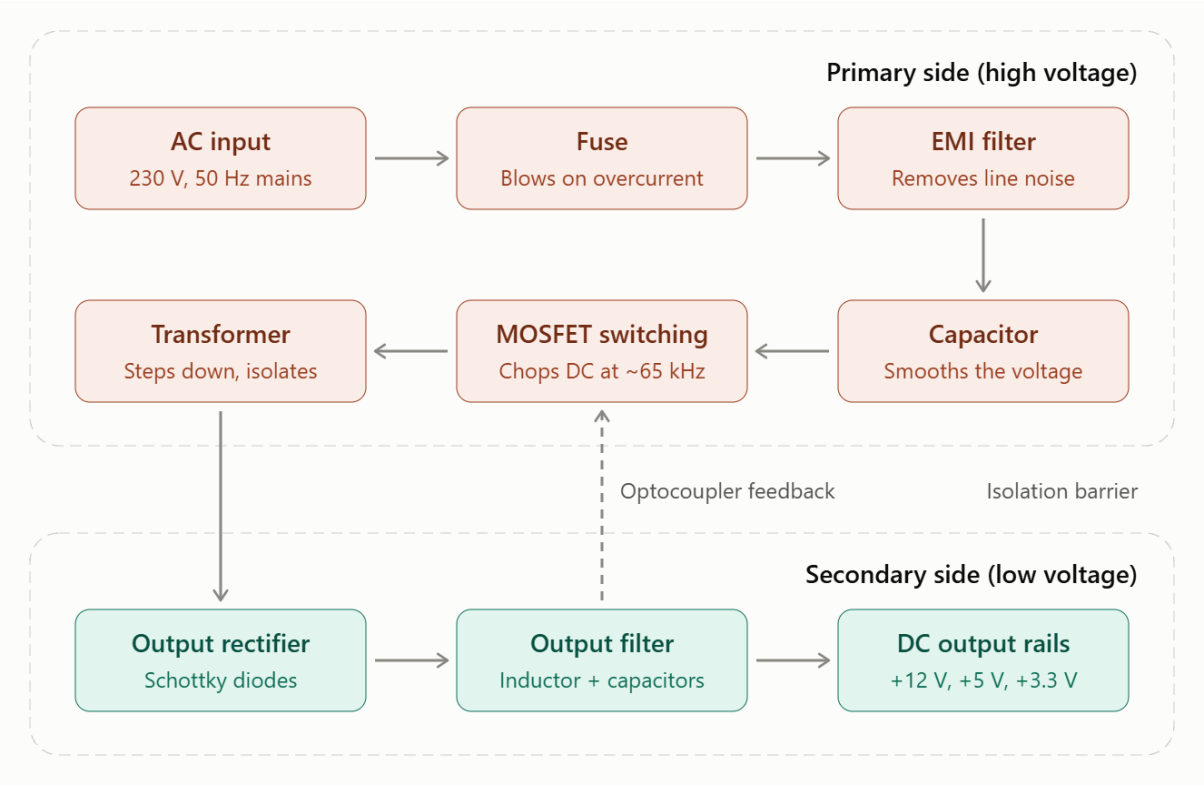
Three Devices That Use SMPS

No.	Device	Explanation
1	Desktop Computer	The SMPS converts AC electricity from the wall socket into different DC voltages required by the motherboard, CPU, GPU, storage drives, and other components.
2	Laptop Charger	The charger uses an SMPS circuit to convert AC power into suitable DC power for charging the laptop battery and powering the laptop.
3	Mobile Phone Charger	The charger converts high-voltage AC electricity into low-voltage DC power suitable for safely charging a mobile phone.

Two Devices That Use UPS

No.	Device	Explanation
1	Wi-Fi Router	A UPS provides temporary backup power during a power cut, allowing the router to keep working if the internet service is still available.
2	CCTV Security System	A UPS supplies backup power to security cameras and recording equipment during a power failure, helping maintain security monitoring.

Task 2 : Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.



Working of SMPS (Switched-Mode Power Supply)

1. AC Input: The SMPS receives 230V AC electricity from the wall socket.

2. Fuse and EMI Filter: The fuse protects the circuit from excessive current, while the EMI filter reduces electrical noise.

3. Rectifier and Capacitor: The rectifier converts AC into DC, and the capacitor smooths the voltage.

4. MOSFET Switching: The MOSFET switches the DC at high frequency to help control power conversion.

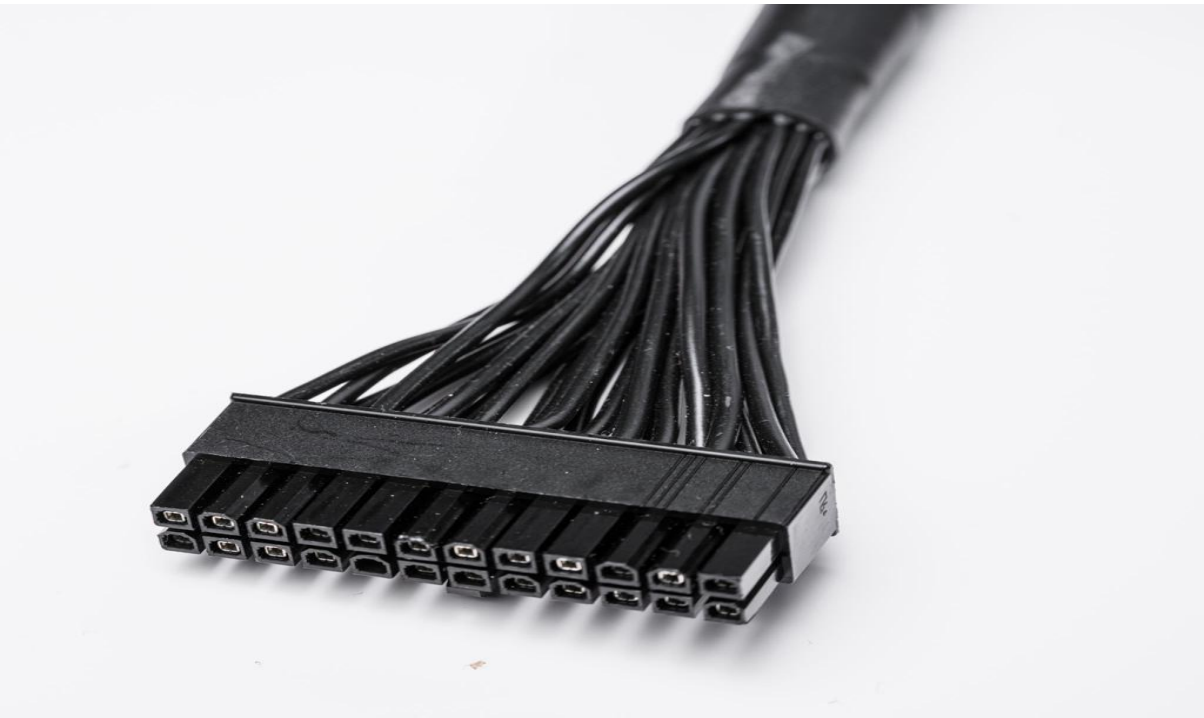
5. Transformer: The transformer changes the voltage to the required level and provides electrical isolation between the primary and secondary sides.

6. Output Rectifier and Filter: These components convert the transformer's high-frequency output into smooth DC power.

7. DC Output: The SMPS provides different DC voltages, such as +12V, +5V and +3.3V, to the motherboard, CPU, storage drives and other computer components.

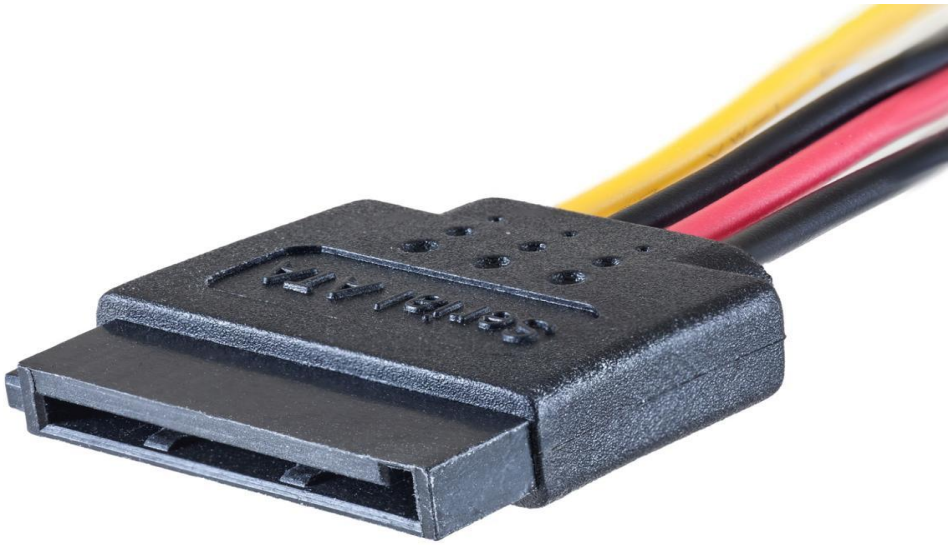
Task 3 : Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

1. ATX 24-pin Motherboard Power Connector



1. **Name**: ATX 24-pin motherboard power connector.
2. **Devices powered**: Supplies the main power to the motherboard, allowing its circuits and connected components to operate.
3. **Safety tip**: Align the connector correctly with the socket and press gently until the locking clip clicks into place.

2. SATA Power Connector



1. **Name**: SATA power connector.
2. **Devices powered**: SATA hard disk drives (HDDs), SATA solid-state drives (SSDs), and optical drives.
3. **Safety tip**: Match the L-shaped connector with the socket. Never force it into place.

3. Molex 4-pin Peripheral Power Connector



1. **Name**: Molex 4-pin peripheral power connector.
2. **Devices powered**: Older hard drives, optical drives, and some cabinet fans.
3. **Safety tip**: Hold the connector body while unplugging it. Never pull the wires, as they may get damaged.

4. PCIe 6/8-pin Graphics Card Power Connector



1. **Name**: PCIe 6/8-pin graphics card power connector.
2. **Devices powered**: Supplies additional power to compatible dedicated graphics cards (GPUs) for graphics processing.
3. **Safety tip**: Use the correct PCIe power cable and ensure it is securely connected to the graphics card.

Task 4 : Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

Connecting the SMPS in a Gaming PC

An SMPS (Switched-Mode Power Supply) supplies electrical power to the motherboard, CPU, graphics card, storage devices, and other components. The following steps explain how to connect it properly when building a new gaming PC.

Step 1: Prepare the SMPS

Switch off the SMPS and unplug the AC power cable from the wall socket. Install the SMPS inside the PC cabinet and secure it with screws. Identify the required power cables.

Step 2: Connect the Motherboard

Use the 24-pin ATX power connector from the SMPS. Connect it to the large 24-pin socket on the motherboard. It supplies the main power required by the motherboard and its circuits. Press gently until the locking clip is secured.

Step 3: Connect the CPU

Use the 4-pin or 8-pin EPS/CPU power connector. Connect it to the CPU power socket, usually located near the processor at the top of the motherboard. This cable supplies power to the CPU through the motherboard's voltage-regulation circuits.

Important: Do not confuse the CPU/EPS cable with the PCIe graphics card cable.

Step 4: Connect the Graphics Card (GPU)

Use the PCIe 6-pin, 8-pin (6+2-pin), or 12V-2x6 power connector, depending on the graphics card model. Connect it to the GPU's power socket. This provides the additional power needed for gaming, rendering, and other graphics-intensive tasks.

Make sure the connector is fully inserted and its locking clip is secure.

Step 5: Connect Storage Devices

- SATA HDD or SSD: Connect a SATA power cable from the SMPS to the drive's power socket.
- M.2 NVMe SSD: It normally receives power directly from the motherboard and does not require a separate SMPS power cable.
- SATA optical drive: If installed, connect a SATA power cable to supply electricity to the drive.

Step 6: Connect Cabinet Fans and Accessories

Use a SATA power connector or 4-pin Molex connector when required by compatible cabinet fans, fan hubs, or RGB controllers. Some fans connect directly to motherboard fan headers instead, so follow the device's instructions.

Step 7: Manage the Cables

Arrange the power cables neatly behind the motherboard tray or in designated cable-management spaces. Keep cables away from fan blades and avoid sharply bending them. Good cable management improves airflow and makes maintenance easier.

Step 8: Check Connections and Start the PC

Before switching on the PC, check the following:

- The 24-pin ATX cable is secure.
- The CPU/EPS power cable is connected.
- The graphics card's required power cable is connected.

- SATA power cables are connected to the storage devices.
- No cables are blocking the fans.

Close the cabinet, connect the AC power cable, switch on the SMPS, and start the PC. Check that the system starts correctly.

Component	**Cable/connector used**
Motherboard	24-pin ATX
CPU power	4-pin or 8-pin EPS/CPU
Graphics card	PCIe 6/8-pin or 12V-2x6
SATA HDD/SSD	SATA power
M.2 NVMe SSD	No separate SMPS cable
Compatible fans/accessories	SATA power or Molex, if required